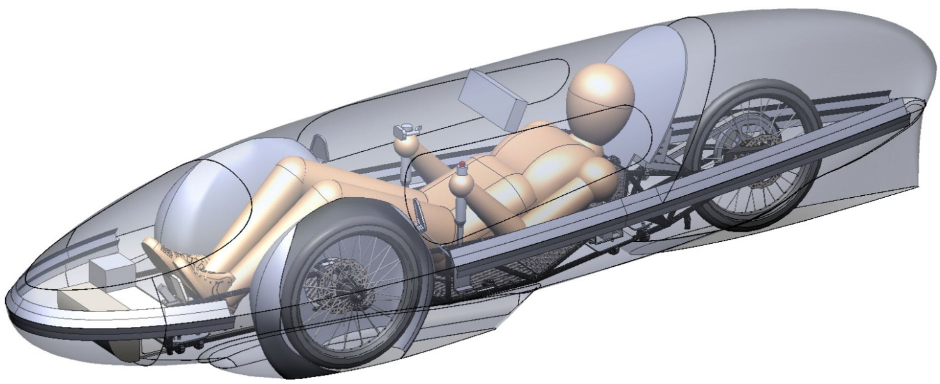
Imperial College London

Department of Aeronautics

Group Design Project

Shell Eco Marathon

Track Simulation, Race Strategy & Cruise Control



Coordinator: Professor Joaquim Peiró
Supervisor: Dr. George Papadakis
Academic year: 2020/2021
Student: Zhi Hang Zhang
CID: 01347531
Date: 14/06/2021

Abstract

The Shell Eco-Marathon is an annual competition organised for students to design, test and race the most energy-efficient vehicle. The 2021 Aeronautics GDP was tasked to improve upon previous developments for the competition in 2022. This included, but was not limited to, the physical design and ergonomics, as well as non-physical aspects such as track simulation and racing strategies.

The track simulation, race strategy and cruise control team was instructed to improve the current IRG lap simulator done by previous projects. In previous editions of the simulator, the results were generated through a collection of Simulink models and MATLAB scripts. In this iteration, we have integrated the multiple subsystems into one combined Simulink model. In addition, the team designed and incorporated a visual tool that displays all numeric data into an animated dashboard, representing the trajectory of the car's path around the track. The improved simulator provided us with a more useful and holistic tool to measure the performance of the car throughout the race.

After obtaining an initial test result from the new simulator, further work was done to optimise the performance of the car by utilising various strategic options at different stages of the race. This race strategy section looked at investigating the effects of varying race-specific parameters (e.g. racing lines, acceleration and cornering strategies) on the overall performance. Other physical aspects that are inherent only to the car design (e.g. mass and drag of the car) were done in a separate sensitivity analysis by other team members.

As the technical consultant for the project, the team made use of the simulator and results to give suitable and justified recommendations to IRG. The options and justifications are analysed extensively and presented in this report. In general, the team recommends the use of a constant speed approach, with no coasting and overtaking, as well as a nominal current for initial acceleration to achieve the best energy performance. Using these recommendations, the performance index from the simulator measures at 1060.68 km/kWh, 17.33% shy of the current world record set by Duke University in 2019 at the Shell Eco-Marathon Americas in California.

Contents

List of Figures

List of Tables

1 Introduction

The Imperial Racing Green (IRG) plans to participate in the Shell Eco-Marathon Europe in Summer 2022, at the former Brooklands Circuit (now Mercedes Benz World) in Weybridge, Surrey. The objective of the competition is to determine the most energy-efficient car that covers a race distance of 11 laps (15620 km) under 39 minutes.

The Track Simulation and Strategy Functional Area Technical Team (FATT) was tasked to upgrade the existing IRG lap simulator developed by previous teams (2020 Aeronautics GDP and 2021 Mechanical Engineering DMT), through cooperation with Powertrain and Vehicle Dynamics teams to estimate, quantify, and report on the performance of the car, before giving suitable recommendations to IRG. The design parameters of the car are listed in the executive summary of the project (Appendix A).

2 Appraisal of Previous Designs of IRG Lap Simulator

2.1 2020 Aeronautics Group Design Project (GDP)

The lap simulator developed by Nicholas Tham [1] from the 2020 Aeronautics GDP involved the collective use of MATLAB scripts from different technical areas (e.g. motor, powertrain) to estimate the performance of the car. The simulator could predict and produce the velocity profile, total energy consumed, performance index, and the completion time of the car. Based on their preliminary results, they deduced that the optimal strategy was to accelerate and maintain a constant speed at straight paths and reduce speed in corners. As this iteration did not have any visual tools, our team believed that having one would better aid the user and IRG in terms of clarity about the car's performance around the track. This simulator was also inaccurate due to the lack of corner performance modelling.

2.2 2021 Mechanical Engineering Design, Make & Test (DMT) Project

The lap simulator from the 2020 GDP was transferred to the 2021 DMT to make further improvements. Thomas Williams from the DMT project made use of Simulink and Simscape models to create a visual flowchart of the simulator. This Simulink model provided relevant linkage between the electrical systems of the motor to the mechanical aspects of the drivetrain (Appendix B). We realised the importance and clarity of having a visual flowchart, hence the team inherited the visual framework as the basis for the improved simulator.

A few problems arose when this simulator was used. As the group inherited the strategy from the 2020 GDP, this resulted in irregularities: Mechanical power was not accurate, and there was a lack of sensitivity of mass on the performance of the car. The DMT and IRG agreed that the presence of inaccurate behaviour in the system stemmed from the very short time spent on optimising the model.

2.3 Improvement to Modules in the Lap Simulator

Following previous designs, the following areas were identified as potential improvements to the lap simulator:

1. Track module: Previous designs did not research on the track and corner profiles extensively, resulting in an idealistic estimation of the performance. The track module seeks to explore the corners, elevation, and positional profiles of the track.

2. Vehicle module: This module involves the integration of several subsystems, such as motor, vehicle dynamics, drivetrain, and cruise control system for the car to measure and gauge the overall performance of the car.

3. Envirionmental module: This describes the different weather conditions, as well as the surface of the track (e.g. Wind speeds and directions, ambient and track temperatures, sunlight intensity and dry and wet surfaces). It also aims to estimate the performance of the car under these environmental conditions.

4. Traffic module: The traffic module was developed to simulate the effects and visualisation of other dummy vehicles, as well to investigate the effect of traffic on the race. The vehicles are assumed to have similar energy-efficient driving strategies, hence they will have similar acceleration and velocity compared to our car.

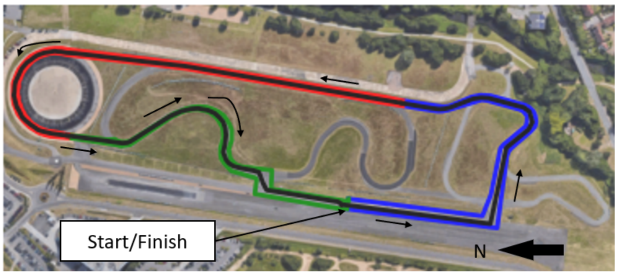
5. Strategic considerations: This involves the creation and analysis of suitable strategies that affect the overall performance of the car throughout the race. More strategies were devised in this iteration to give a more holistic understanding of the range of strategies available to the team.

3 Design of the Lap Simulator

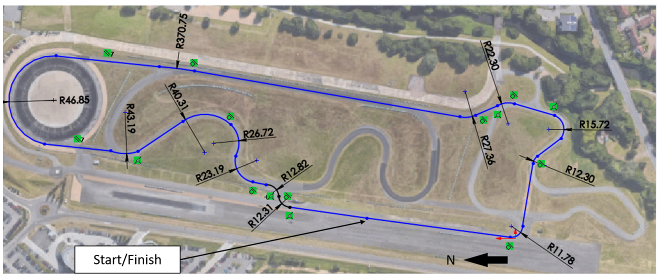
3.1 Track Module

3.1.1 Track Profiling

The objective of the track module was to provide the necessary information about the track profiles and its corners, and also as a visual tool to the performance of the car on the racetrack. The Brooklands Circuit (now the Mercedes-Benz World) in Weybridge, Surrey was aerially scouted using Google Maps to determine the track and corner profiles, shown in Figure 1.



(a) Brooklands Circuit in Surrey with anticlockwise race path.



(b) Midline corner profiles of the Brooklands Circuit.

Figure 1: Track profile of Brooklands Circuit.

Once the coordinates of the track were obtained, they were then discretised into small segments of approximately 1m and this was useful in determining the position of the car on the racetrack.

From online data [2], the terrain of the track is generally flat, with an elevation height between 12 and 17 metres above mean sea level. The gradient of the track is very gentle, with little height variations throughout. Hence, we assumed that that the effects of elevation were negligible in the simulations.

3.1.2 Racing Lines

In typical speed-racing terminology, the racing line is the path that would allow the car to clear the track in the quickest possible time [3]. However, while the competition has a time limit, it is not the objective to complete it fast. Hence, the racing lines we choose should seek to optimise the energy consumption of the car. We have decided to select 3 lines: the Midline (MRL), Ideal (IRL) and Efficient (ERL). The description and justification of the 3 lines are detailed in Table 1:

Racing line	Description	Number of turns	Lap length / m
Midline (MRL)	Midline dimensions of the racetrack	13	1420
Ideal (IRL)	Shortest line through corners	14	1404
Efficient (ERL)	Most efficient line for the motor (Discussed in **Section 3.2.2**)	14	1432

Table 1: Description of chosen racing lines. The MRL was chosen as the benchmark across the 3 lines as it is the most intuitive and simple line to choose on the racetrack.

3.2 Vehicle Module

The vehicle module has the following subsystems integrated into the simulator: motor, vehicle dynamics, drivetrain, electrical systems and cruise control. The functions of each subsystem are described below.

3.2.1 Motor System

The central component to the car is the motor systems, which provide a platform of translating electrical energy input to mechanical energy for movement. The team had to first learn about the inner workings of the motor, before coordinating with the Powertrain team on the range of requirements needed for a suitable motor selection. Please refer to the report done by *Adrian Czuchaj & Gonzalo Diaz-Pavon* for more details about the motor selection.

From our initial estimations, we determined that a suitable motor would need to have a high electrical-to-mechanical efficiency as well as good torque performance. This would contribute to an optimal car performance during the race.

3.2.2 Vehicle Dynamics

The vehicle dynamics portion involved the study of analysing the equations of motion in the car, factoring in various forces that contribute to the car's velocity and movement. This included the contributions of rolling resistance, aerodynamic drag as well as cornering forces. Please refer to the report done by *Parco Wong* for more details about vehicle dynamics.

Upon further analysis into the turning performance of the car, we found out that the work done by the car is significantly higher when turning into a sharper corner, as shown in Figure 2. This is an inherent vehicle property and is, more importantly, independent of our motor choice.

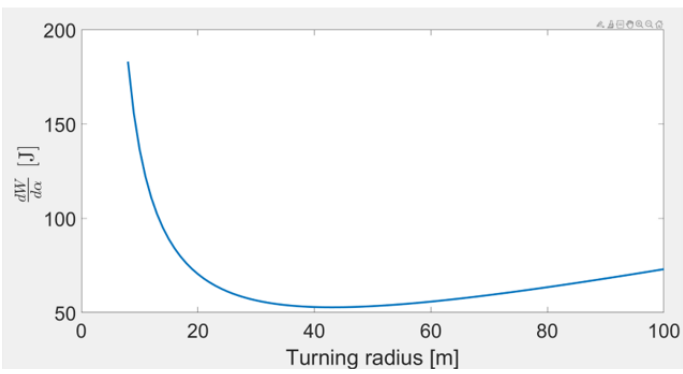


Figure 2: Relation between work done and turning radius.

This means that if we could keep a turning radius of a corner between the optimal values of 40 - 50 m, the work done by the car would be reduced, resulting in a lower energy consumption. This was the main motivation to look at a new racing line that explored having turns as close to the optimal values as much as possible, and hence gave rise to the ERL as listed in Table 1.

3.2.3 Drivetrain

The linkage between the motor and the vehicle is the drivetrain. From the previous projects, a planetary gearhead with a chain drive was implemented to amplify the torque from the motor to the driving wheel. With the information of the motor and the car, the Powertrain team could then reformulate the optimal drivetrain to ensure greater efficiency between the motor and the car. This was also an iterative process as we could also analyse a range of suitable motors and check which one provided the most optimal performance. Please refer to the report done by *Reem Shaltout* for more details about drivetrain justifications.

3.2.4 Electrical Systems

From the 2021 DMT project, the team attempted to model the battery and DC-DC converter into the model, but did not manage to accomplish the desired effect. The team then sought to improve upon that to achieve the same aim. The incorporation of electrical systems also allow us to monitor the battery cell pack, such as state of charge, discharge current, and voltage. Please refer to the report done by *Jiaxuan Tang* for more details about the design of the battery cells.

3.2.5 Cruise Control

As IRG was not employing a regenerative braking system, a cruise control system was developed to allow the car to travel at a constant speed around the track, while accounting for uneven surfaces and increased cornering forces. Please refer to the report done by *Thomas Brown* for more details about the design of the cruise control system.

3.3 Features of the 2021 GDP Lap Simulator

Taking into account of all the modules, the FATT decided to develop 2 separate models:
(a) Integrated Simulink model (Figure 3 and Appendix B): An all-in-one model detailing interactions between the electrical, motor, powertrain, and dynamics systems.
(b) Visual tool (Figure 4): A newly-designed animated dashboard displaying car position on the racetrack, as well as important parameters such as energy and cornering performance.

These 2 models are run sequentially: Model **(a)** calculates the parameters such as velocity and energy consumption over the course of the race. Subsequently, post-processing is done to estimate the cornering performance of the car. Random traffic is then generated before the model **(b)** is activated to display the parameters and plots.

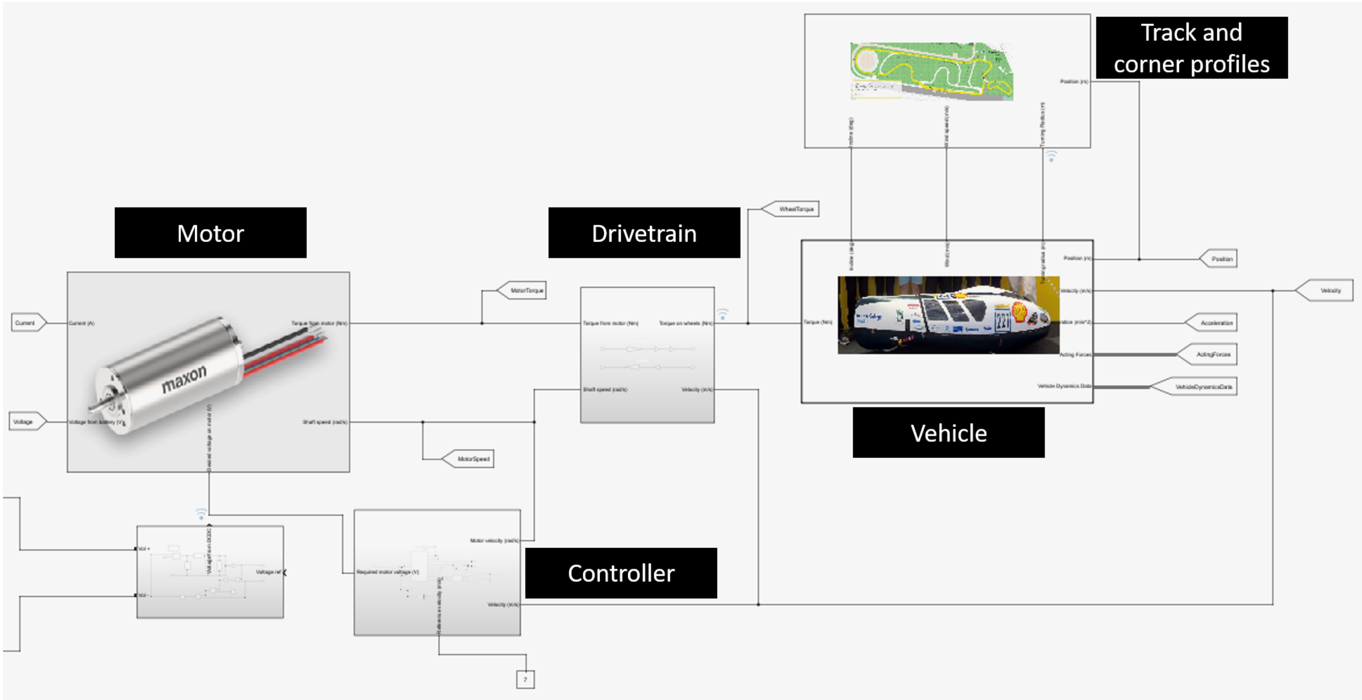


Figure 3: Schematic of integrated systems in the simulator, excluding the battery subsystem.

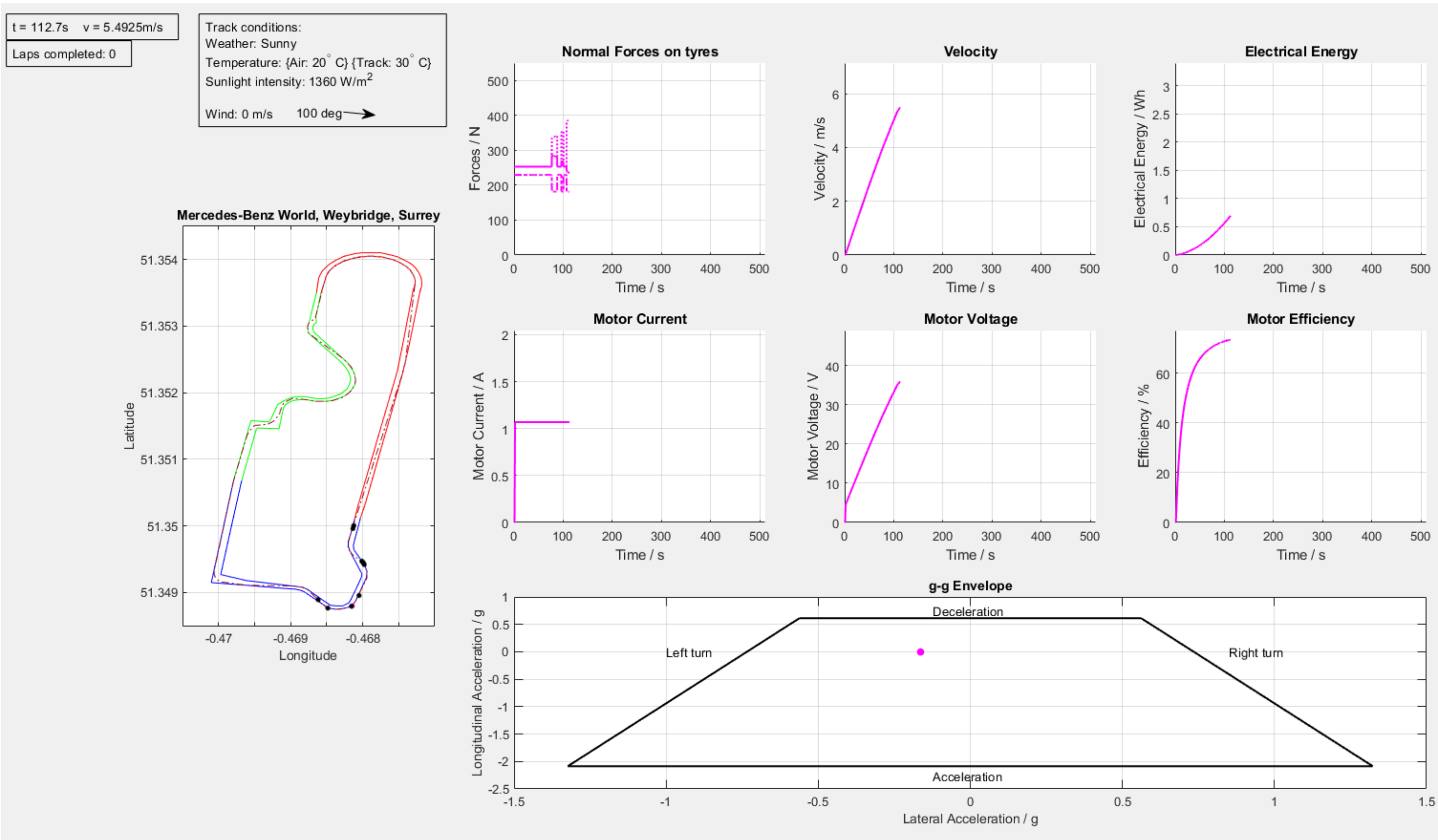


Figure 4: Visual tool display, showcasing animated dashboard of position and parameters.

4 Race Strategies

The race strategies during the competition are arguably as important as the design of the car in determining the performance of the car. Employing suitable strategies throughout different phases of the race could bring about significant benefits to the energy performance. In this section, comparative studies of different strategies at different stages of the race are discussed extensively.

4.1 Summary of Recommended Race Strategies

Table 2 summarises the different recommended race strategies to adopt during the race:

Race stage	**Recommended strategy**
Cruise speed?	Set a target cruise speed of 7 m/s.
Constant speed or Constant power?	Cruise control with constant speed, with PID controller.
Racing line?	Adopting IRL and/or ERL.
Initial acceleration from rest?	Supply constant nominal current to the motor until it reaches constant speed.
Driving & Corner strategy?	No coasting at corners, maintain constant cruise speed throughout the race. Coasting only possible at the final stretch to the completion of the race.
Overtaking?	Avoid overtaking at all as much as possible; can choose to stagger start time away from main traffic. Focus and commit to one fixed strategy.

Table 2: Summary of recommended race strategies at different stages of the race.

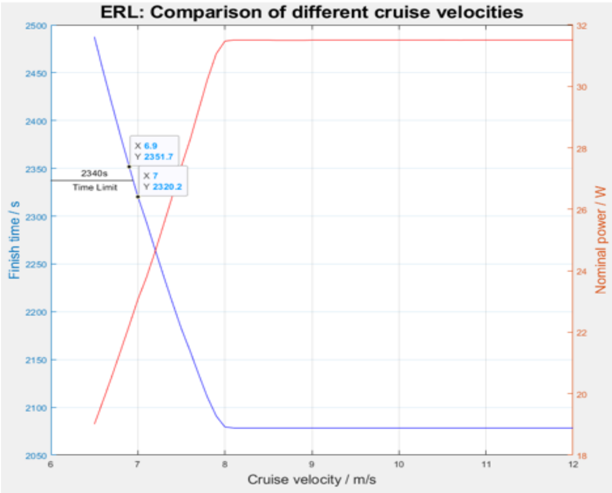
These strategies were implemented into the main simulator. After employing all the recommendations, the best achievable performance index was **1060.68 km/kWh**. The justifications to these strategies are explained in the following sections.

4.2 Cruising Speed

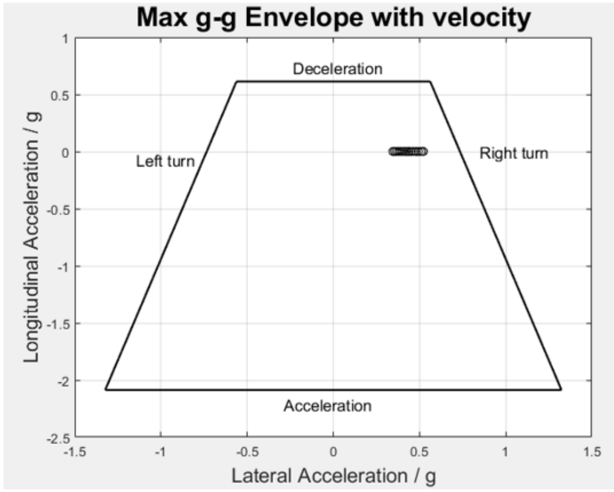
One of the main constraints of the race is that the 11 laps should be completed within 39 minutes (2340 seconds). A starting simple calculation would give an average speed of 6.68 m/s throughout the race from start to finish. Clearly, this is unrealistic if the initial acceleration and necessary deceleration during corners are factored in.

Hence, a range of target constant cruising speeds were explored and simulated to determine the optimal cruise speed. The velocity profile of the car would follow an initial acceleration and cruise at the constant speed until the end of the race.

Across the choice of racing lines, we realised that a constant speed of 7 m/s was the most suitable and realistic cruise speed. Figure 5 shows an example of the analysis when applied to the ERL:



(a) Comparison of cruise velocities with estimated finish times and nominal power.



(b) Maximum g-g Envelope of different cruise velocities.

Figure 5: Comparative studies of cruise velocity on the ERL.

A velocity of 7 m/s meets the time limit, has the best energy performance and satisfies cornering stability concerns. Subsequent analysis done in the simulator is based on this value of velocity, and **our team recommends IRG to select a 7 m/s cruise velocity** throughout the race.

4.3 Constant Speed vs Constant Power

As IRG was not considering a regenerative system, the necessity of reducing speed at corners was greatly reduced. Hence, a cruise control system was implemented to ensure the car is going at the selected cruise velocity, and establish the efficient use of the motor. A constant speed approach involves a PID control, while a constant power requires a constant voltage input to the motor.

Our findings indicate that using constant power was beneficial in initial testing, where cornering forces and bumps on the track were not modelled extensively. However, this approach was flawed once all the models were integrated together, and was unable to keep the velocity to the desired value especially at corners. On the other hand, PID control is very robust and flexible; the ability to control the velocity fairly accurately while accounting for random factors and cornering made it an **ideal cruise control application** in the simulator. Figure 6 shows that the PID controller still manages to keep the velocity close to its target even on a bumpy track. This demonstrates the robustness, effectiveness and flexibility of PID controller on cruise control.

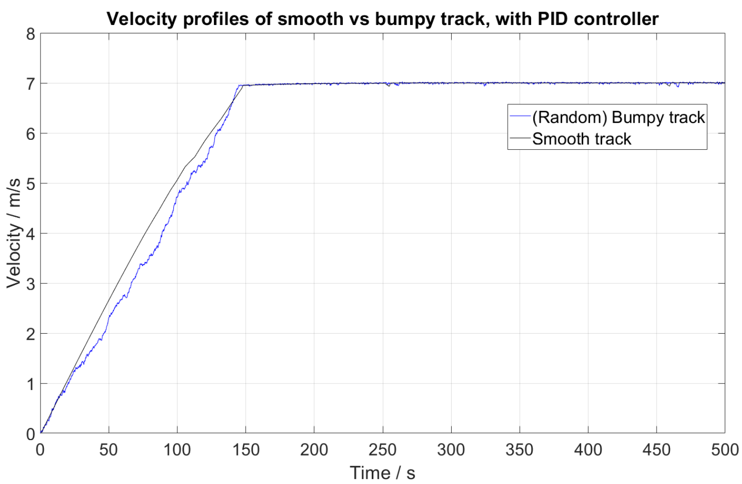


Figure 6: Velocity profile across different track scenarios with a target velocity of 7 m/s.

4.4 Racing Lines

The 3 trajectories described in Section 3.1.2 were analysed to determine the final completion times, electrical energy consumption, nominal power and performance. The results are tabulated in Table 3:

Racing line	**Average speed / m/s**	**Estimated completion time / s**	**Average motor efficiency / %**	**Nominal Power / W**	**Performance index / km/kWh**	**Difference in performance index / %**
MRL *(Benchmark)*	6.75	2305.8 (38 mins 25.8s)	72.91	27.02	899.63	-
IRL	6.75	2286.7 (38 mins 6.7s)	73.46	23.00	1057.50	+17.55
ERL	6.76	2330.6 (38 mins 50.6s)	73.79	22.94	1060.68	+17.90

Table 3: Comparative study of racing lines on car performance.

It is evidently clear that the IRL and ERL outperformed the MRL despite the vastly different characteristics between them. The IRL scored well due to its shorter lap distance, while the ERL's advantage stems from a more efficient use of its motor despite covering more ground. The ERL, however, carries the risk of not completing the race in time, especially if insufficient buffer is allocated for model inaccuracies, driver error or the presence of heavy concentrated traffic on the track.

Hence, **the FATT recommends using both IRL and ERL**, interchangeably if neccesary, as its default racing lines during the race. This added flexibility could prove to be advantageous when the car encounters traffic.

4.5 Initial Acceleration

The speed of the initial start-up acceleration from rest was also investigated. A slow acceleration will reduce the energy usage but puts a time pressure on the car to complete the race. This will prompt the need for higher average speeds to make up for lost time, which can hurt the energy performance as seen in Section 4.2. Hence, a fast acceleration was set as the default in the simulator.

The amount of torque supplied to the car depends on the current supplied to the motor. Hence, for a fast acceleration (a high rate of current influx), there was a further choice to be made on how much electrical current to supply to the motor.

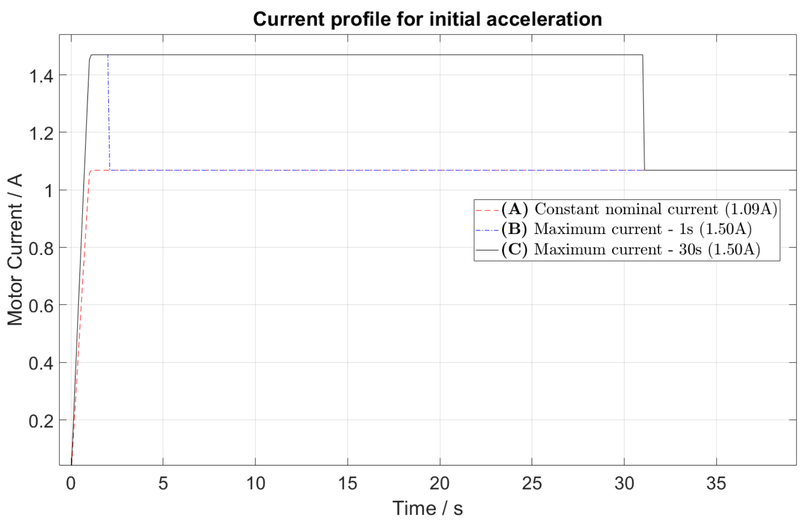


Figure 7: Comparison of current spike profiles for initial acceleration.

The current supplied can be split into 2 approaches (Figure 7):
(A): Constant nominal (maximum continuous) current at 1.09 A, or
(B & C): Constant maximum short-time operational current at 1.50 A for a fixed time, before supplying constant nominal current.

With maximum current, the car is able to start cruising earlier, and therefore requires less time to finish the race. However, this presents operational risks, as the short-time operational current cannot be supplied for sustained periods of time to prevent overheating and damage to the motor.

The limit of short-time operations were not specified by the manufacturers, hence more testing should be done to determine the maximum time the motor is able to sustain high currents. In our analysis, we compared supplying maximum current to the motor for 1 second **(B)** and for 30 seconds **(C)**.

The results of the different current profiles when applied to the ERL are tabulated in Table 4:

Acceleration strategy	**Time to cruise / s**	**Estimated completion time / s**	**Average motor efficiency / %**	**Nominal Power / W**	**Performance index / km/kWh**	**Difference in performance index / %**
(A) Nominal current	167.0	2330.6 (38 mins 50.6s)	73.79	22.94	1060.68	-
(B) Maximum current (1 s)	166.3	2329.9 (38 mins 49.9s)	73.78	22.95	1060.31	-0.0349
(C) Maximum current (30 s)	153.3	2318.7 (38 mins 38.7s)	73.70	23.09	1058.96	-0.162

Table 4: Comparative results of current profiles for initial acceleration.

From Table 4, the performance of the car was further reduced when the duration of maximum current is increased from 1 to 30 seconds, while only slightly improving completion time. With the results, we conclude that the amount of current supplied during initial acceleration has a knock-on effect on the overall performance of the car. Hence, we would **recommend to use the nominal current approach (A)** during initial acceleration from rest, especially if the limits of the motor are untested.

4.6 Driving & Corner Strategy

One of the crucial aspects of the optimal strategy in previous projects was reducing speed with no throttle input (coasting) on the entry of a corner, before accelerating and maintaining constant speed on the straights. With the improved simulator, we sought to reevaluate and compare the relevance and performance between the previous (coasting) (α) strategy and a constant cruise speed (β) strategy. As there is no regenerative braking in the system, the supplied voltage is assumed to be zero when the throttle is disengaged in our simulator.

Our team picked 3 main parameters for comparison: Energy consumed, Average motor efficiency and Power for (re)acceleration. To minimise computation time while still obtaining reliable results, we chose the sample time to be 450 s, in which the car should cover about 20 turns. This is sufficient to give an indication of the performance of the driving and cornering strategies. The comparative study results are displayed in Table 5.

<table>
<tr><th rowspan="2">Driving strategy</th><th rowspan="2">Energy consumption in 450 s / Wh</th><th colspan="2">Average motor efficiency / %</th><th rowspan="2">Power consumed during acceleration / W</th></tr>
<tr><th>Peak</th><th>At 450 s</th></tr>
<tr><td>(α) Coasting & reacceleration</td><td>3.79</td><td>65.12 (113.0 s)</td><td>36.00</td><td>40.39</td></tr>
<tr><td>(β) Constant cruise speed</td><td>2.89</td><td colspan="2">72.17</td><td>31.04</td></tr>
</table>

Table 5: Comparative results of driving strategies for 450 seconds.

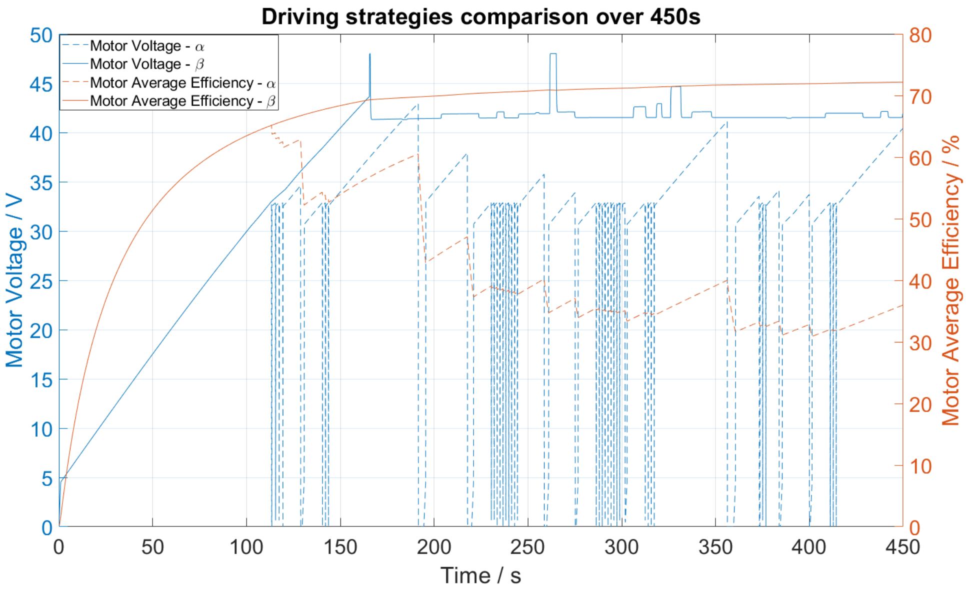


Figure 8: Trend of efficiency and voltage profile across α and β strategies.

From Figure 8, there are massive disadvantages by using strategy α. It incurs a higher energy consumption and sharp drop in motor efficiency. The reduced voltage input also results in a sharp deceleration, hence it is only applicable once the car has reached an acceptable velocity. Reacceleration is also expensive; each attempt to accelerate the car after a corner would penalise the energy performance of the car. Hence, the coasting strategy was ineffective to achieve the energy-efficient aim of the race, and should be avoided as far as possible.

Furthermore, with no environmental factors, the maximum cornering speed of the car is 9 m/s. Hence, the current cruise speed of 7 m/s can be maintained with no stability concerns, which gives even lesser incentive for the car to coast. However, coasting may be needed should the maximum cornering speed be lower than the target cruise speed, in light of stability concerns.

The cons of coasting are further compounded by the absence of regenerative braking of the system, where there is no avenue to channel the generated energy from the motor into the batteries. A preliminary simulation of regenerative braking during coasting, with 20 V input at idle throttle, would potentially reduce the energy consumption of strategy α by 4.90% over 450 s.

However, coasting could be applied when the car is on the last stretch of the race, provided there is sufficient remaining time. As described by previous strategic studies, the car would switch off its motor at the last stretch of the last lap, and use the momentum of the car to coast through the finish line. Using rough estimates, the car decelerates at 0.37 m/s^2, which gives a distance of 66.22m to cover. This point coincides with the start of the main straight leading towards the finishing line, thus the driver could switch off the motor after turning the last corner. Coasting at the last stretch can improve the performance index of the ERL by 0.24% (1063.21 km/kWh). However, it would require approximately 19 extra seconds to coast to the finish line, hence the driver would need to plan their time strategically.

The team understands that due to the added complexity of the regenerative braking system, IRG opted to shelve the idea and proceeded with the cruise control model instead. Without a regenerative braking system, the **team recommends against the use of a coasting strategy** and agrees with a cruise control over the duration and length of the race, with coasting only possible at the last stretch of the race. Further studies to determine when coasting should begin should be conducted to generate a more comprehensive strategy.

4.7 Overtaking

IRG had requested the FATT to look into overtaking as part of their arsenal of strategies, looking into the strategic concerns about releasing or racing in heavy traffic conditions.

The Shell Eco-Marathon places significant emphasis on the efficiency of the vehicle, much less about completing the race quickly. Previous entrants and world-record holders have pushed the time to the limits to obtain their desired efficiency.

From Section 4.6, we have established clearly that acceleration is disadvantageous to the energy performance of the car, and should therefore be restricted to initial stages when speeding up to cruise speed. Hence, it can be deduced that accelerating to overtake even one vehicle is akin to reacceleration in strategy α, and therefore would be detrimental to the performance.

Due to vehicle stability concerns, overtaking at corners is highly discouraged as there is an maximum speed the car can corner at. Failing to meet this limitation would result in poor energy performance or worse, potential damage to the car. According to the rules of the Shell Eco-Marathon, the driver of the overtaking car takes full responsibility of the overtaking manoeuvre [4], hence any driver error at corners will directly affect its result.

Furthermore, in our analysis of the ERL, we have utilised wider lines at sharper corners, thus the probability of overtaking at corners is significantly reduced as the car will need to travel longer distances than other vehicles. In the event of traffic, we recommend IRG to switch between IRL and ERL for navigation without compromising overall performance.

Every team is also allowed to choose when to start the race, hence the timings can be staggered out between one another to ensure traffic is avoided. It is a lose-lose situation for most, if not all vehicles if they need to execute overtaking manoeuvres by acceleration.

To achieve high efficiency and good performance, most competitors will be travelling around the same velocity (between 6.7 - 8 m/s (see Table 6)) throughout the race; even if there is a need to overtake, the car should be able to execute it without the need to change its speed.

4.8 Performance comparison and evaluation

We compared our performance index with similar competitors or previous designs to evaluate our performance. Table 6 shows the comparison of the performance index estimates:

Project	**Performance Index / km/kWh**	**Cruise speed / m/s**
2020 Aeronautics GDP [1]	1381.60	7.00
2021 Mechanical Engineering DMT	267.11	7.12
2019 Duke University *Eta* [5] *(2019 Guinness World Record)*	*1283.00*	6.67
2018 Duke University *Maxwell* [5] *(2018 Guinness World Record)*	*1164.00*	6.94
2021 Aeronautics GDP	1060.68	6.76

Table 6: Comparison of performance index of similar vehicles. Values in *italics* are actual values measured at Shell Eco-Marathon tournaments.

4.9 Discussion

We improved upon the 2020 GDP by taking into account improved drivetrain and motor changes, as well as cornering performance. We subsequently used the simulator to generate the results with the recommended strategies, and they were comparable to other cars travelling at similar cruise speeds.

The 2020 Aeronautics GDP acknowledged that there was an overestimation of the performance index due to the lack of cornering mechanics and realistic racing lines and strategies. For the 2021 Mechanical Engineering DMT, the significantly poorer performance index was due to an older and inefficient motor being used to facilitate their testing process, resulting in an expected, though unsatisfactory, performance.

However, there were still several factors unaccounted for that may reduce the accuracy of the simulator. The current model, though addressing the issues faced in the 2020 GDP, still does not model extensively the efficiency of the controller, actual motor efficiency, the effects of the environment (e.g. winds, temperature) and potential effects of losses in the electronics system. The efficacy of the performance also depends on specific driving scenarios, such as the amount of buffer time set by IRG and accounting for potential driver errors.

4.10 Further Improvements

Certain improvements can be made to improve the simulator. The most important development should involve research into potential losses of the various models: controller efficiency losses and actual motor performance. These are more likely to be achieved by regularly testing the various components to obtain realible results for use. More work can also be devoted to quantifying environmental impacts on the performance, as most of the efforts for this iteration were focused on the modelling of the motor, drivetrain and vehicle modules. On-site track reconnaissance could also be done to determine a more accurate track profile and cornering data.

It would be extremely beneficial if IRG could do a physical test run of the current car design, or obtain extensive information from other competitors, to validate the results from the simulator. This is an extremely important step as the simulator needs to give convergent results with actual test runs, so that inconsistencies and further enhancements can be studied and rectified. This is also a common practice employed by the racing community, with established F1 teams like McLaren dedicating manhours into the simulator to test, validate and simulate their operational race cars during a race weekend in order to guarantee a strong performance [6].

Lastly, a probable future project could bring this simulator to run in real time, for example by using a Graphic User Interface (GUI). This will allow the team and/or driver to select the preferred racing lines and make decisions according to real-time movement and traffic. The team will get the opportunity to learn how to react to situations in real time, and drivers can get the necessary training to cope and learn with car controls and analyse their performance across a variety of scenarios.

5 Conclusion

In conclusion, the 2021 Aeronautics GDP has made significant progress in the development of the lap simulator. The FATT prioritised the modelling of the motor and vehicle dynamics to give more accurate estimates of the car performance. This is also the first iteration where a visual tool for the car around the racetrack was developed, and the dashboard was a convenient way to display all important parameters during the race.

The team also explored more strategies than previous projects, diving into more detail about the possible strategies throughout different stages of the race. These recommendations made to IRG should bring out the best performance in the car based on its current design and specifications.

References

[1] Tham N. *Shell Eco-Marathon, Vehicle Dynamics & Traction: Lap Simalator and Strategy Development*. Imperial College London.

[2] FreeMapTools.com *Elevation Finder*. Available from: https://www.freemaptools.com/elevation-finder.htm [Accessed 12 May 2021].

[3] Bayerische Motoren Werke AG. *12 pro tips: How to find the racing line on any corner*. Available from: https://www.bmw.com/en/performance/how-to-find-the-racing-line.html [Accessed 12 June 2021].

[4] Shell Eco-Marathon. *Shell Eco-Marathon: 2021 Official Rules, Chapter 1*. Available from: https://base.makethefuture.shell/en_gb/service/api/home/shell-eco-marathon/global-rules/_jcr_content/root/content/document_listing/items/download.stream/1598906823204/311003400ef3284afdeb403265a7b6001dfb9b7b/shell-eco-marathon-2021-official-rules-chapter-I.pdf [Accessed 12 June 2021].

[5] Duke Electric Vehicles. *Prototype Evolution*. Available from: http://www.duke-ev.org/prototype-evolution [Accessed 12 June 2021].

[6] Formula 1. *The Key Role Of The Sim Driver In F1*. Available from: https://www.youtube.com/watch?v=7FcgmarQNLc [Accessed 12 June 2021].

A Appendix - Executive Summary

A.1 Project Coordination, Communication and Rules Compliance

The project coordination involved managing 26 students working remotely, subdivided into functional-area technical teams. A project management Gantt chart was made public, with tasks, internal deadlines and weekly design reviews. The allocation of tasks to team members was managed accordingly to relative importance and difficulty. Scrum meetings were held near every morning for group discussion of big picture progress, internal feedback and design decisions. Weekly design reviews with the client Imperial Racing Green (IRG) and academic supervisors provided external feedback. The final design recommendations describe a highly efficient prototype with improved performance, manufacturability and operational usability.
For more information on 'Project Coordination', please refer to the report by Adrien Legris

In order to be deemed safe and legal to compete, the design must comply with Shell Eco-Marathon Rules and Regulations. It must also fit within the constraints of the customer, in terms of both the budget and also manufacturing capabilities. The consequences for not adhering to these constrains range from additional expenditure to complete disqualification from the event hence it is key to ensure design compliance. Changes in the design process were closely monitored and controlled throughout the project in order to maintain design compliance. In particular, care was taken when one sub-team made a change to a part of the design which also affected another sub-team. The process for ensuring the design was complaint was first to give each sub-team their specific rules and then to monitor any changes and communicate them to the relevant parties. The end design was fully complaint to the relevant rules at this stage of the process.
For more information on 'Rules Compliance', please refer to the report by Khristian Delizo-Salabit

A.2 Packaging, Integration and CAD

The CAD, Packaging and Integration team handled any changes to the design of the car using SolidWorks 2020 and used CAD to aid in the design of components of the SEM car. With a bird's-eye view of the project, the sub-team was responsible for ensuring that the work carried out by different FATTs formed a cohesive and efficient design. This meant working on key aspects of the car, such as packaging of the energy and propulsion modules in the rear of the chassis and obtaining a suitable mannequin model to visualise the placement of components within the car. The CAD sub-team also took a crucial role in the design of the steering mechanism and Aeroshell packaging, due to the complex nature of these components. Hence, the sub-team's work entailed keeping the CAD of the car organised and up-to-date, as well as communicating changes to the FATTs whose designs might be affected by other changes made to the vehicle. This sub-team is also responsible for creating the documentation such as the presentation, the poster and video.
For more information on 'Packaging, Integration and CAD', please refer to the following reports: Dominic Heslop, Shi Zhang Neo, Uddeshya Saini

A.3 Vehicle Chassis

The CFRP spaceframe chassis is designed as an upgrade and revised chassis from the DMT Group 26 chassis. The chassis will accommodate the driver and various components of the vehicle. The design is verified with Steering, Vehicle Dynamics, Ergonomics, Packaging and Powertrain to ensure the vehicle achieves the desired performance and fulfills the SEM rules and regulations.
The space frame will be made of CFRP tubes with joints of ABS lug-inserts along with wet-layup carbon fibre clothes, which is a strong and lightweight joint that is easily manufactured.
Multiple changes are applied to the previous chassis, such as increasing the number of bars in the chassis structure, adding a front section and optimising the space of the driver's compartment. The

static and dynamic load cases are identified and tested on the structure. A 3D FEA model with beam elements was then created to observe the details of axial forces, moments, deformations and rotations more accurately. Beam members withstanding less forces and moments are optimised in terms of diameter and thickness for weight and cost reduction.
The final chassis structure gives a maximum deflection of 1.92 mm under static loading and 1.74 degree rotation in the aggressive cornering loading and 0.72 degree rotation in entry loading , which are within the requirements of the IRG team.
For more information on the 'Chassis' design, please refer to the following reports: Cheuk Lam Lam, Alec Yiu

A.4 Aeroshell Aerodynamics

The aerodynamics team had to design a shell that would be aerodynamically competitive, comply with the competition rules, client requirements, and meet the packaging constraints. Specifically, IRG set a target of 20% reduction in drag and a neutral lift coefficient.
Previous SEM-winning aeroshell designs were initially considered. Our main goal was to make the design as streamlined as possible. A suitable top view aerofoil that meets the packaging constraints and ensures rear visibility was chosen - the aerofoil has a near-tear drop shape which is known to be ideal. An aerofoil could not fit the side view profile due to the flat bottom part, hence elliptical shapes were used between the packaging constraints points. We tried to make the side view profile as slender as possible but the height restrictions didn't allow that in the rear to reduce wake drag to a great extent but the result is definitely of acceptable standard given the constraints.
The CFD analysis of the aeroshell was carried out in STAR-CCM+ and involved first validating CFD settings versus Ahmed body experimental data. Next, a mesh convergence study was then carried out on the first iteration of the aeroshell to determine a suitable cell count at which to run design iteration simulations. The validation process yielded a 6.5% error for a grid convergence index of 2.9%. The final aeroshell design as given in the CAD of our SEM car had a C_D of 0.084 and a C_DA of 0.03090. This a respectable 50% and 48% improvement on the C_D and C_DA respectively over the aeroshell that IRG is currently using.
For more information on 'Aeroshell Aerodynamics', please refer to the following reports: Jonathan De Sousa, Harry Taylor-Shaw, Stelios Theocharous

A.5 Aeroshell Structural Design

The aeroshell must resist a set of loads expected during operational use.These were established and quantified, the most critical being lifting and transportation, as well as driver exit. This, combined with manufacturability and cost constraints, influenced the decision to construct the aeroshell using CFRP. Following this, an FEA model was validated, and executed on the aeroshell, indicating that a [0,90,+45,-45] symmetric layup, using a Carbon Fibre-Epoxy UD Pre-preg was sufficient to resist the loads identified by the team.
Joints across the aeroshell were acknowledged as a key detriment to the stiffness of the vehicle. Thus, the number of detachable panels featured in the final design has been minimised to 3 (being manufactured as 5). A key connection occurs between the upper and lower halves of the aeroshell, for which a lightweight rim ($\sim$ 1.1 kg) has been designed, revolving around the free edge perimeter of the sections, secured using velcro. This maximises the surface area of the joints (promoting the transfer of load), whilst assisting in the alignment of the two halves.
The Aeroshell-Chassis joint was flagged by the client as an area for development. In response, the final design places 4 CRFP tubes at corners of the chassis, which concentric tubes mounted on the aeroshell underside slot into. This solution, whilst constraining all degrees of freedom, only requires 4 M3 wingnuts to fasten, promoting the ease at which the aeroshell underside panel can be removed for maintenance.

Finally, to satisfy regulations, a crumple zone was designed, featuring a foam block optimised to maximise impact duration. A sacrificial nose cone has been introduced into the panel configuration, which has been positioned to prevent damage propagation to the remainder of the aeroshell, lessening potential repair costs.
For more information on the 'Aeroshell Structural' Design, please refer to the following reports: Achal Srivastav, Alex Choy and Piragash Selvakumar.

A.6 Powertrain

Our responsibility was to revise the motor, gearhead, the drive train and the tyre and wheel selection. For the motor selection, we needed a motor that operates on a nominal voltage of 48V or less so that it would be compatible with the current battery. A shortlist of motors were tested using a motor simulator to compare a range of key parameters; as a result, the Maxon ECX TORQUE 22 XL (48 V version) was selected. This motor gave us an optimal gear ratio of 3.87 for the gear-head selection, so we selected the Ceramic GPX 22 as it has a gear ratio of 3.9 and is compatible with our motor selection. A motor mount was designed to carry and secure both the motor and the gear-head whilst allowing us to align the motor and for us to tension the chain. The drive train was kept as the previous design of a chain drive and sprockets, so an appropriate chain guard was designed. For wheel and tyre selection, rolling resistance was the most critical parameter. Given the weight and operating speed of the vehicle, and considering the budget, bicycle wheels were selected. Research was conducted into which bicycle tyres offered the least rolling resistance and it was found that for a manageable cost penalty, the Michelin Prototype 45/75 R16 were chosen for their unbeatable coefficient of rolling resistance of 0.00081. The remainder of the wheel was chosen to be of a hub and spoke design as the low cost was far more favourable than the minimal weight savings that would come from a fully covered aero-blade CFRP wheel. An off the shelf wheel rim was chosen and modified to accommodate the specific rim profile requirement of the tyre.
For more information on the 'Powertrain' design, please refer to the following reports: Gonzalo Diaz-Pavon Lo, Reem Shaltout and Will Lugo.

A.7 Steering and Braking

The steering subteam is responsible for designing of a new steering system. A simpler side steering mechanism with single lever input and linkage steering replaces the old steering-wheel pinion-and-rack system. The new mechanism performs better in controllability and maintainabiliy, and also benefits the ergonomics design. Based on a understeer characteristic of the system, studies were done to re-modify the steering geometry where the balance is set for more power efficiency than cornering performance while ensuring stability.
The main goals of the braking team was to re-design a lightweight foot pedal assembly and to choose a rear braking system so that the braking safety test imposed by SEM could be passed. The new foot pedal assembly was re-designed and is made of Aluminium 6082-T6 and ABS-M30. The new foot pedal assembly is 33% lighter than the previous one but further weight reduction is possible. For the rear braking system, the Shimano Deore 10s series was chosen for its reliability and low cost.
For more information on the 'Steering and Braking' design, please refer to the following reports: Jiaxin Zheng and Alexandre Donat

A.8 Driver Environment, Ergonomics, and Interfaces

This subteam was tasked with improving the overall comfort of the driver. This was done by reducing cockpit temperatures and humidity, improving ventilation, ensuring adequate visibility, revising the driver's seating position and reducing driver fatigue.
To ensure required visibility, the driver's field of vision was studied at the most constraining corners

of the track. Rear-view mirrors were added at appropriate positions in cases of being overtaken. Driver's seating position has been redesigned and included vibration dampening gel cushions to ensure comfort and visibility throughout the race. Interfaces such as steering, dashboard and throttle were redesigned and adjusted to reduce driver fatigue. To improve the driver environment, the aeroshell was painted white for heat reflection, window area was reduced, ventilation holes were incorporated into the aeroshell and the driver was equipped with a cooling vest.
For more information on the 'Driver Environment, Ergonomics, and Interfaces' design, please refer to the following reports: Aoi Izumi, Joshua Tong

A.9 Dashboard, Communications & Sensors Study

The roles of this team were to design the dashboard, communication module and horn. The dashboard displays the important vehicle working parameters and warnings for the driver in a clear and convenient way. The communication module sends data from the sensors to the team. This helps the team know the present conditions of the vehicle and driver. The communication module should ensure that the team in any position of the race receive the data accurately and rapidly. The driver may use the horn when overtaking other contestants or in an emergency situation.
For more information on 'Dashboard and Communications', please refer to the following reports: Zhecheng Shan

Sensor study focuses on different sensors which can be applied to this vehicle and other electronics which were not mentioned. The target of the sensors is to improve the safety and monitor the efficiency. To achieve this, a number of sensors are used and the detailed products with specifications are given including acceleration sensor MPU6050, current sensor DHAB S-134 and temperature sensor DHT22. The cost, power and performance are evaluated to shortlist. Position of the sensors and other electronics are included to facilitate the sensing and the layout of the cables. Detailed connections and cable routing are specified to ensure good ergonomics and no interference will occur. Simulink models of electronics were also built to assist the Track Simulation Team while it takes significant time to run. Improvement should be implemented on these models. Further test should be done on this part as hardware programming is needed, which cannot be achieved especially in connection and compatibility.
For more information on the 'Sensor Study' design, please refer to the following reports: Jiaxuan Tang

A.10 Vehicle Dynamics and Traction

The role of vehicle dynamic is to analyse the performance and stability of the vehicle, as well as to maximise efficiency by recommending an optimal parameter across sub-teams. The main focus is on the tyre selections, CG position, load distribution, wheelbase, tyre drag and vehicle handling.
Two models are developed to determine the tyre load distribution and the system response. The first initial model is restricted by a constant speed of 7 m/s (most efficient), and the mid-line turning radius is applied for calculations. Even though the model is not practically feasible, it was utilised at the early stage of the project to give a sensible approximation. The maximum steering angle and aligning moment are estimated for the steering team when designing the optimal steering geometry. The impacts on wheelbase, track and CG position are reviewed and a design limitation envelope is generated for vehicle stability. Crosswind condition is also taken into account alongside the aerodynamic team's data, resulting in an additional 26 N acting on the aerodynamic center. Optimisation of vehicle handling is achieved by comparing the QSS response and the transient response in all turns. Furthermore, the steady-state surface map with the input domain provides complete information on the cornering performance, in which speed and steering angles are no longer constant within the analysis. Both models are integrated into the simulator to implement live feedback on the vehicle performance. A stability check has also been done via the simulator and is concluded to be

stable.
For more information on the 'Vehicle dynamics and traction' design, please refer to the following reports: Parco Pak Long Wong

A.11 Track Simulation, Race Strategy and Cruise Control

The track simulation team's task was to develop the lap simulator allowing accurate performance prediction of car designs while implementing and testing different racing strategies. The software was created by incorporating the models of other FATTs such as powertrain, vehicle dynamics and electrical systems into one combined, functioning model. A mapped trajectory was created to represent the track and recognise the states of the vehicle. The team then performed a sensitivity analysis on crucial parameters of the car to assess their performance and identify further improvements. To assist data inspection, the team implemented the visual tools displaying the position and parameters of the car around the racetrack enabling investigation of the cornering performance and various range of strategies employed by the user. Throughout the project, the team also developed two tools, a simplified benchmark model used by other FATTs in the conceptual design and a motor efficiency mapping tool supporting the motor choice process.
The race strategy team was responsible for coming up with suitable strategies for the race. These strategies had to be efficient-centric and relevant comparative studies were made to optimise the performance of the car. The cruise control team was largely responsible for determining the controller needed to maintain a constant speed during the race.
For more information on the 'Track Simulation, Race Strategy and Cruise Control', please refer to the following reports: Adrian Czuchaj, Zhi Hang Zhang and Thomas Brown

Component	Desciption
Aeroshell	5 panel CFRP composite shell, with 3 windows.
Wheels & Steering	Bicycle wheels, Michelin Prototype 45/75 R16 , indirect lever-actuated steering
Chassis	Composite tube spaceframe, ABS lugs with carbon fibre wetlayup
Motor	ECX Torque 22 XL 55W brushless motor
Battery	26 Samsung 25R cells enclosed in an acrylic case
Safety Equipment	Driver harness, disk brakes, mirrors foam impact structure, energy compartment bulkhead, roll-bar
Electronics	NUCLEO-L432KC MCU, DC-DC converter, ESCON70/10 Motor controller, Sensors

Table 7: Description of main components.

Parameter	Value
Wheelbase / Track Width	1425mm / 530mm
Mass	31.8kg
Ground Clearance	85mm
Length / Height	2922mm / 662mm
CdA	0.031
Rolling Resistance coefficient	0.001
Project Cost	£7,720
Energy Usage	14.85 kWh/run
Performance Estimation	1060.68 km/kWh

Table 8: Table of key parameters.

B Appendix - 2021 Lap Simulator

B.1 2021 Mechanical Engineering DMT Project

Figure 9 shows the Simulink and Simscape model of the 2021 DMT project.

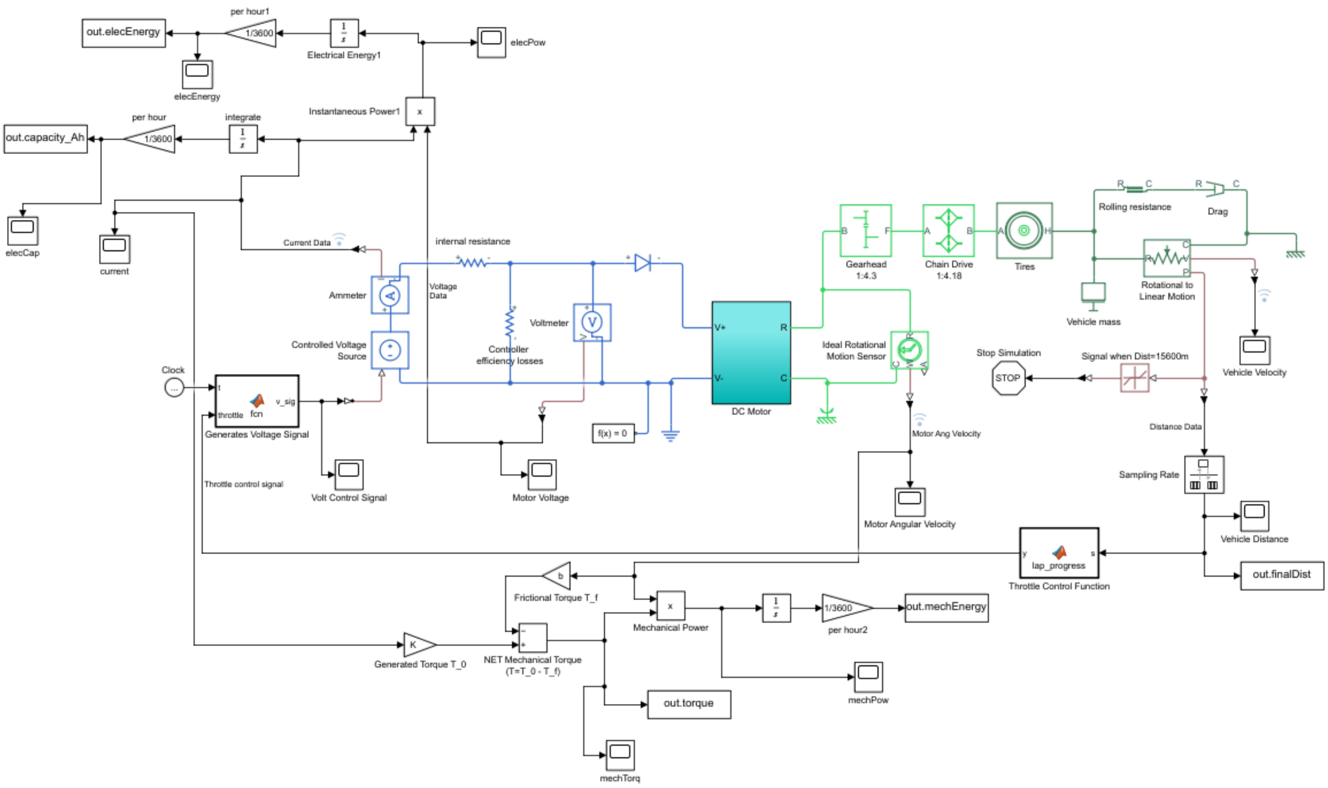
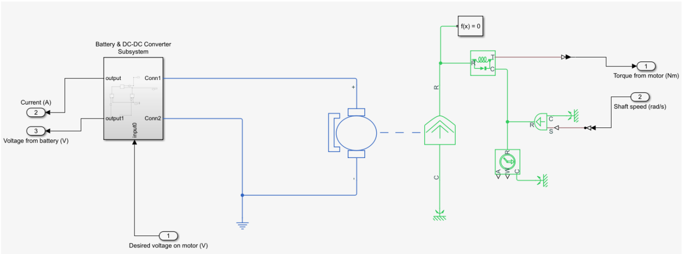


Figure 9: Simulink model of 2021 DMT project.

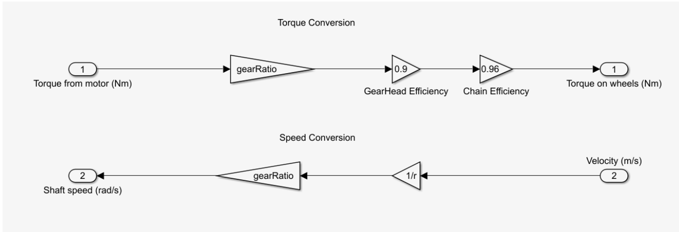
The visual representation of the simulator proved to very useful and convenient. Hence, the framework was adopted for the 2021 GDP Lap Simulator.

B.2 2021 Lap Simulator Simulink Integrated Model

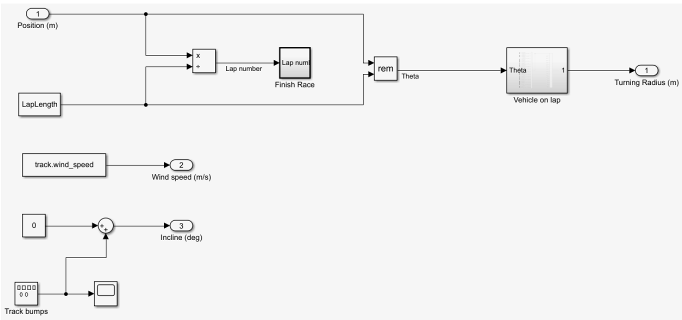
Figure 10 showcases the various components of the improved lap simulator.



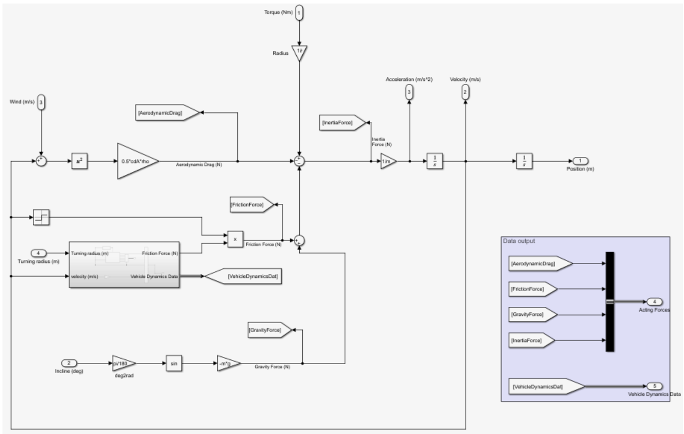
(a) Motor subsystem. An equivalent circuit block was used to model the selected brushless DC motor.



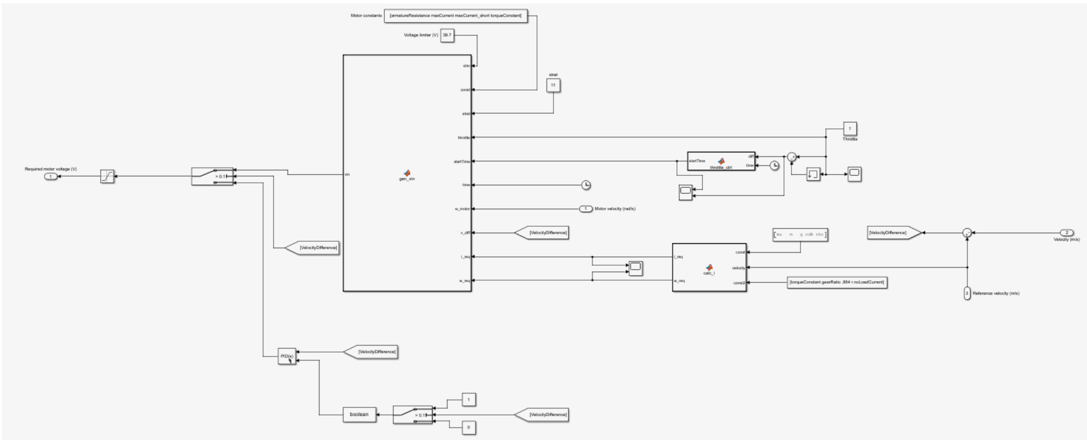
(b) Drivetrain subsystem. This consists of the implemented planetary gearhead, as well as the chain sprockets.



(c) Track subsystem. This subsystem was used to calculate the position of the car on the racetrack, as well as to factor in cornering performance.



(d) Vehicle subsystem. This module calculated all the forces acting on the car to produce values for acceleration and velocity.



(e) Controller subsystem. This controlled the required voltage input into the motor to produce the required torque to accelerate the car. The PID controller is also placed here to control the velocity when it reaches cruising velocity.

Figure 10: Detailed breakdown of the 2021 lap simulator.